slides ⌥⌘7

Kubernetes workshop

2023 Frameworks Onsite

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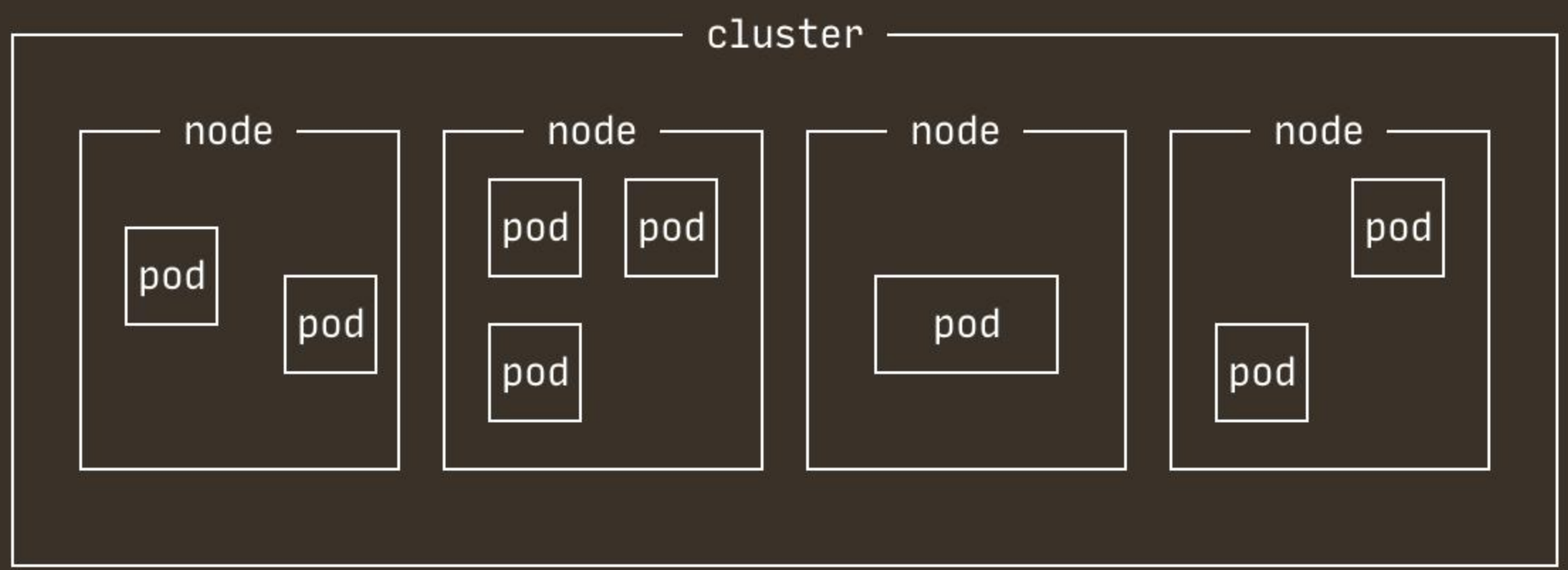
Architecture refresher

Architecture refresher: Kubernetes clusters

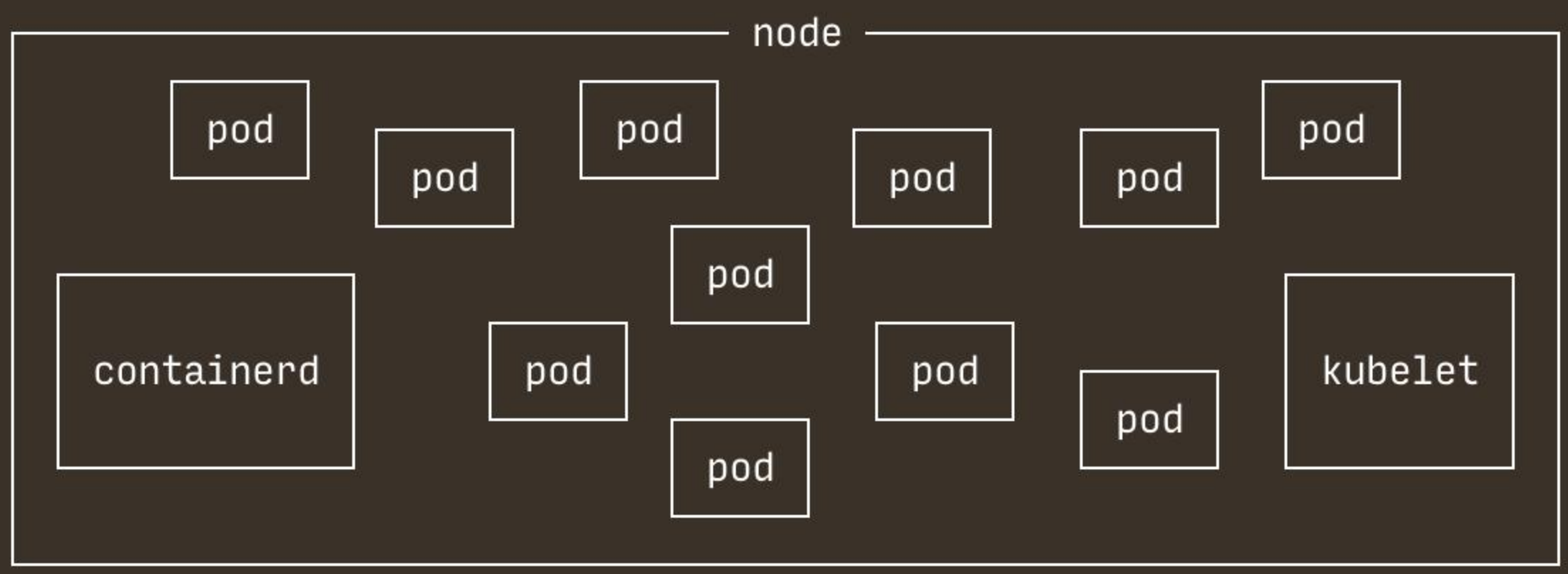


Architecture refresher: Kubernetes clusters

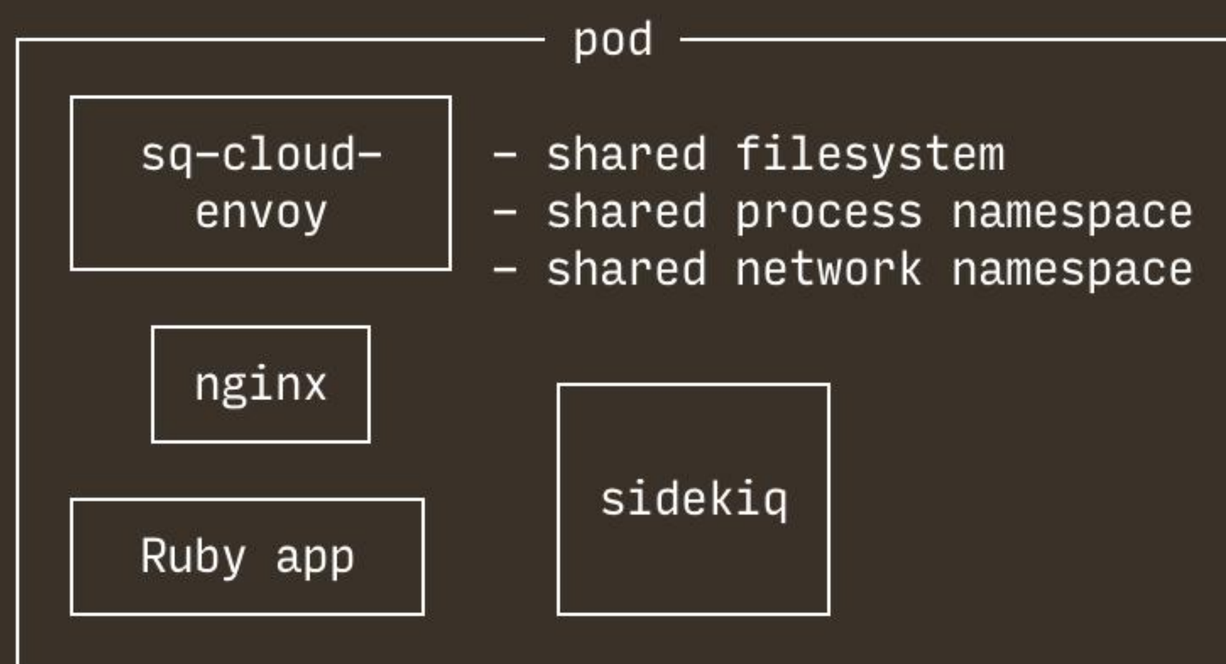
Examining locally configured SKI clusters...

```
kubectl config view | yq '{"clusters": [.clusters[].name]}'
```

Architecture refresher: Kubernetes nodes



Architecture refresher: Kubernetes pods



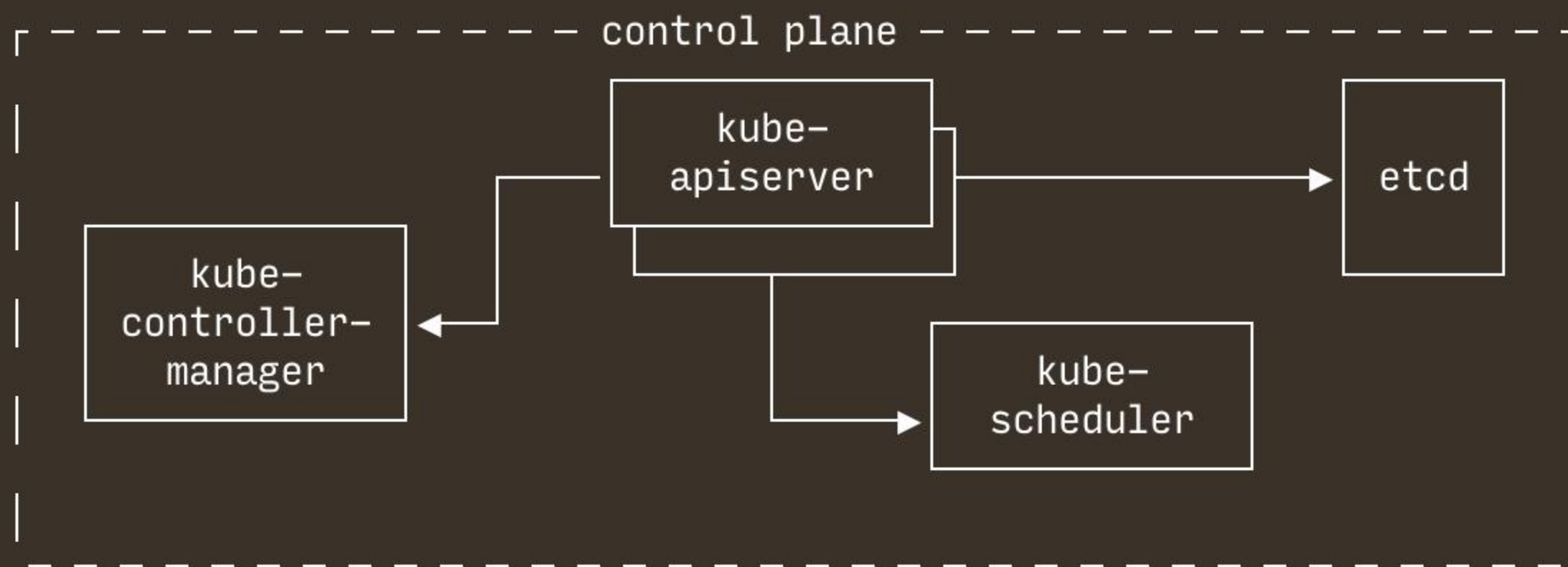
- shared filesystem
- shared process namespace
- shared network namespace

Architecture refresher: Kubernetes containers

```
container

- image
- command (ENTRYPOINT)
- args (CMD)
- environment variables
- resource requests
- resource limits
- readiness probes
- liveness probes
- ...
```

Architecture refresher: Kubernetes control plane



Workloads

Workloads

Overview

- Pod
- Deployment and ReplicaSet
- StatefulSet
- DaemonSet
- Job and CronJob

Workloads: Deployment vs. ReplicaSet

Deployment

- Describes the desired state of the workload
- Defines the update strategy for achieving the desired state
- Creates ReplicaSets and manages how pods are allocated between them

Workloads: Deployment vs. ReplicaSet

ReplicaSet

- Ensures a given number of pods are running
- Lower-level concept than a Deployment
- One potential use case: custom update orchestration

Debugging

Debugging

Where am I?

```
sq ski context current
```

Debugging

Where am I?

```
kubectl config current-context
```

Debugging

Getting to where you want to be

```
sq ski context switch --help
```

Debugging

Getting to where you want to be

```
kubectl config set-context --help
```

Debugging

Don't have access?

- https://go/registry OR
- https://go/aws

Debugging

Don't have access?

```
sq ski context update --help
```

Debugging

Don't have access?

```
beyond-curl -s \
https://registry.sqprod.co/api/v1/user/ntarrh/aws_credentials \
| head -n 7
```

Debugging

Don't have access?

```
aws-creds get --help
```

Debugging

Don't have access?

```
aws-creds get 212513346311 read-only | jq '.'
```

Debugging

Using kubectl

- kubectl VERB NOUN [NAME] [args]
- kubectl get pods
- kubectl get pod NAME -o yaml
- kubectl describe deployment NAME

Debugging

Using kubectl: get vs describe

- kubectl get pod NAME -o yaml
- kubectl describe pod NAME
- helpful: pipe to vim (or your preferred $EDITOR)

Debugging

Logs

- kubectl logs POD [-c CONTAINER]
- kubectl logs POD --all-containers
- kubectl logs POD [-c CONTAINER] --follow
- kubectl logs POD [-c CONTAINER] --previous

Debugging

More logs

```
sq ski debug logs --help
```

Debugging

Attaching a shell

- kubectl exec -it deployment/ski-exemplar [-c CONTAINER] -- /bin/bash

Debugging

Attaching a shell

```
sq ski debug exec --help
```

Debugging

Rolling back

- "Rolling back" means (re)declaring a previous configuration
- Kubernetes doesn't have a rich "history" semantic
- kubectl rollout undo deployment/abc

Debugging

Using k9s

- https://k9scli.io/

Workshop

Prerequisites

- cd ~/Development/topsoil
- ./compost backend OR babushka shared-kube && sq update

Troubleshooting exercise

- sq ski context switch -a ski-exemplar -r west -e staging
- sq ski context switch -a ski-paragon -r west -e staging
- sq ski context switch -a rubyexemplar -r west -e staging